

Y9 5.2-5.3 bestest

Q1

One hundred cards are numbered from 1 to 100. A card is chosen at random. What is the probability that the first digit on the chosen card is 9?

Q2

Ten cards are numbered from 1 to 10. A card is chosen at random. What is the probability that the number on the chosen card is neither even nor a factor of 15.

Q3

A survey of 100 children in a village found that 95 of them had blue eyes. If this is representative of the 3600 people in the whole village, how many blue eyed people could you expect there to be in the village?

Q4

A card is picked at random from a standard deck of 52 cards. Find the probability that it is the ace of hearts.

Q5

A box has 24 marbles that are either green or yellow. Lisa takes one marble out, notes the colour, then puts it back in the box. She does this 40 times.

Colour	Freq.
green	10
yellow	30

What is the relative frequency of green?

Using these results, what is the probability a marble drawn at random will be yellow?

 %

How many green marbles would you expect there to be in the box?

Q6

Musical instruments played.

	guitar (G)	not guitar (G̃)	
saxophone (S)	0.41	0.18	0.59
not saxophone (S̃)	0.15	0.26	0.41
	0.56	0.44	1

What percentage play the saxophone but not the guitar?

 %

What percentage play either the saxophone or guitar?

 %

Find the probability a student chosen at random plays both.

 %

Q7

A box contains one lemon, one raspberry and one cola ice-block. Jeremy chooses two ice-blocks at random.

Draw a tree diagram to help answer the question.

What is the probability the first ice-block is cola?

What is the probability he chose a raspberry followed by a lemon?

What is the probability he chooses a raspberry ice-block?

Q8

In a class of 28 students; 15 study Music, 21 study Visual Arts and 3 study neither.

What is the probability a student chosen at random studies both Music and Visual Arts?

What is the probability a student chosen at random studies Visual Arts but not Music?

What is the probability two students chosen at random both study Music?

Q7

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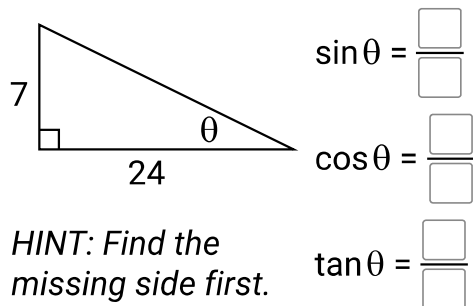
Q8

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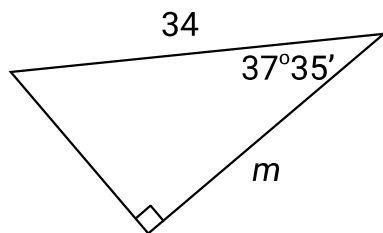
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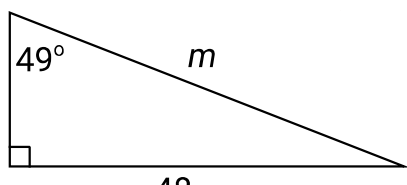
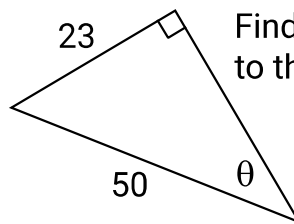
What is the probability two students chosen at random both study Music?

Q22**Q23**

Find the value of m . (1 d.p.)

**Q24**

Find the value of m . (1 d.p.)

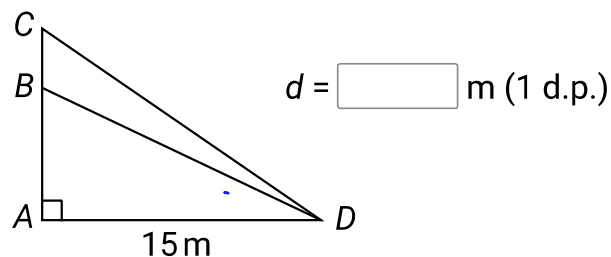
**Q25**

Find the value of θ correct to the nearest degree.

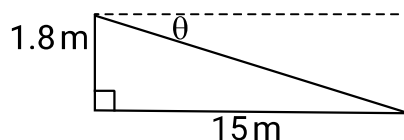
$$\theta = \square^\circ$$

Q26

The angle of elevation from D to B is 25° .
The angle of elevation from D to C is 35° .
 AD is 15 m. Find the distance from B to C .

**Q27**

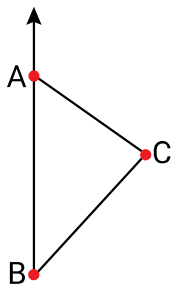
A shark is swimming 15 m from a jetty which is 1.8 m above the water. Find the angle of depression to the shark.



$$\theta = \square^\circ \text{ (nearest degree)}$$

Q28

A is due north of B. C bears 125° from A and 042° from B.



Find the bearing of B from A.

°

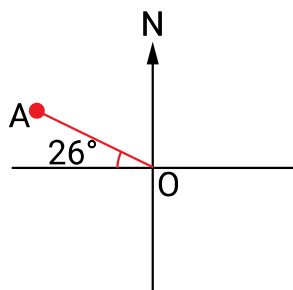
Find the bearing of A from C.

°

Find the bearing of B from C.

°
Q29

Write the 3-figure bearing of A from O.



Bearing = °

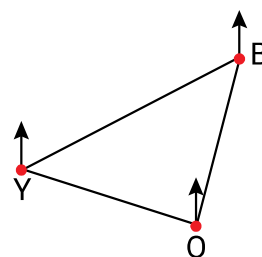
Q30

Max and Luke are standing on opposite sides of a 15m tall flagpole. The angle of elevation from Max to the top of the pole is 23° and from Luke to the top of the pole is 17° . How far apart are Max and Luke?

$d =$ m (1 d.p.)

Q31

O bears 110° from Y. B bears 065° and 023° from Y and O respectively.



$\angle BYO =$ °

$\angle BOY =$ °

$\angle OBY =$ °